newly deployed fixed assets being offset by the decline in depreciation expense from fully depreciated fixed assets.

Gains on Equipment Transactions. We exchanged certain used network equipment and cash consideration for new network equipment resulting in gains of $1.1 million for 2019 and $1.0 million for 2018. The gains are based upon the excess of the estimated fair value of the new network equipment over the carrying amount of the returned used network equipment and the cash paid. The increase in gains from 2018 to 2019 was due to purchasing more equipment under the exchange program in 2019 than we purchased in 2018.

Interest Expense. Interest expense results from interest incurred on our $445.0 million of senior secured notes, interest incurred on our $189.2 million of senior unsecured notes, interest on our installment payment agreement, interest on our finance lease obligations and interest incurred on our €135.0 million of 2024 Notes that we issued on June 25, 2019. Our interest expense increased by 12.5% for 2019 from 2018 primarily due to the issuance of $70.0 million of senior secured notes we issued in August 2018, the issuance of €135.0 million of senior unsecured notes we issued in June 2019 and an increase in our finance lease obligations. The 2024 Notes were issued at par for €135.0 million ($153.7 million) on June 25, 2019. The 2024 Notes were issued in Euros and are reported in our reporting currency — US Dollars. As of December 31, 2019 the 2024 Notes were valued at $151.4 million resulting in an unrealized gain on foreign exchange of $2.3 million in 2019.

Income Tax Expense. Our income tax expense was $15.1 million for 2019 and $12.7 million for 2018. The increase in our income tax expense was primarily related to an increase in our income before income taxes.

Buildings On-net. As of December 31, 2019 and 2018 we had a total of 2,801 and 2,676 on-net buildings connected to our network, respectively.

Year Ended December 31, 2018 Compared to the Year Ended December 31, 2017

Our management reviews and analyzes several key financial measures in order to manage our business and assess the quality of and variability of our service revenue, operating results and cash flows. The following summary tables present a comparison of our results of operations with respect to certain key financial measures. The comparisons illustrated in the tables are discussed in greater detail below.

	Year Ended December 31,		Change
	2018	2017	Percent
	(in thousands)		
Service revenue	$ 520,193	$ 485,175	7.2 %
On−net revenues	374,555	346,445	8.1 %
Off−net revenues	145,004	137,892	5.2 %
Network operations expenses(1)	219,526	209,278	4.9 %
Selling, general, and administrative expenses(2)	133,858	127,915	4.6 %
Depreciation and amortization expenses	81,233	75,926	7.0 %
Gains on equipment transactions	982	3,862	(74.6)%
Interest expense	51,056	48,467	5.3 %
Income tax expense	12,715	25,242	(49.6)%

(1) Includes non-cash equity-based compensation expense of $895 and $604 for 2018 and 2017, respectively.

(2) Includes non-cash equity-based compensation expense of $16,813 and $12,686 for 2018 and 2017, respectively.